

CSS 430 A: Operating Systems

Spring 2021 | Tuesday/Thursday 8:45-10:45AM

Instructor: Dr. Yusuf Pisan, pisan@uw.edu

Office Hours: Monday 1-2pm and Thursday 2-3pm. Signup via [Canvas Calendar](#)

Class Discord: [invite link](#)

Course Description

This course introduces the logical design of operating systems. Topics covered include processes, threads, CPU scheduling, synchronization, deadlocks, memory management, virtual memory, file systems, I/O systems, protection, and security used in the popular desktop and server operating systems. A focus is made on the Unix operating system. The course utilizes Java in programming examples and as a basis for design.

Learning Objectives

ABET Outcomes:

This course contributes to the following ABET outcomes:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, , and economic factors
3. an ability to communicate effectively with a range of audiences

4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Textbooks

Operating System Concepts, <https://www.os-book.com/OS10/> 10th edition, Abraham Silberschatz, Peter Galvin, and Greg Gagne, Addison-Wesley, 2013. (available as an e-text)

Optional References

[*Understanding the Linux Kernel*](#), Daniel P. Bovet and Marco Cesati, O'Reilly, 2000

[*The Design of the Unix Operating System*](#), Maurice J. Bach, Prentice Hall, 1987

[*Advanced Programming in the Unix Environment*](#) 2nd Ed, W. Richard Stevens, Addison Wesley, 2005

Grading

- Exercises: 10%

- Homeworks: 20%
- Projects: 40%
- Midterm: 15%
- Final: 15%

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide. 2.0 roughly corresponds to 75%. Grades are not curved which means everybody can get a 4.0. Your success is not determined by how well or badly other students in the class are doing.

Distance Learning

Owing to the UW response to COVID-19, this course will be delivered entirely online through the Canvas learning management system and Zoom (supplemented by other web resources). This means that you will need to put in more time on your own than in a conventional course. Of course, the benefit of this structure is that you do not need to be physically present at UW-Bothell.

This course is scheduled to run synchronously at your scheduled class time via Zoom. These Zoom class sessions will be recorded. The recording will capture the presenter's audio, video and computer screen. Student audio and video will be recorded if they share their computer audio and video during the recorded session. The recordings will only be accessible to students enrolled in the course to review materials. These recordings will not be shared with or accessible to the public. The University and Zoom have FERPA-compliant agreements in place to protect the security and privacy of UW Zoom accounts. Students who do not wish to be recorded should 1) change their Zoom screen name to hide any personal identifying information such as their name or UW netid, and 2) not share their computer audio or video during their Zoom sessions.

Policies

Pet Policy: If a pet enters the camera frame during class, we will pause our discussion for an introduction to that pet and admiration by all. All pets are welcome.

Assignment Submission: You can miss one exercise without penalty.

For some homeworks and projects, you will be able to work individually or in pairs. See homework/project description on whether pair-work is allowed or not. When working as a pair

- Both students must enter a comment in the Canvas assignment with their name and their partner's name 72 hours before the assignment is due. Submit an empty file if necessary to leave a comment.
- All substantial work must be performed with both students present (virtually).

You can submit one and only one project 24 hours late without any penalty (if working as a pair both people will need to use their one-time extension). If you are using your 24 hour extension, you must post a brief explanation to "Using Extension" assignment before the assignment due date.

If you have any questions about your assignment grade, you must contact me within one week.

Plagiarism: Other than when working in pairs, talking about code is OK, looking at each others code is not OK. Looking at references to understand how a functions gets used is OK; looking up assignment solutions is not OK. It is also not acceptable for your code or solutions to be publicly accessible on the web (for example, a public GitHub repository). Plagiarism will result in an assignment score of zero and a misconduct letter in your student record. Please be very careful to adhere to the student code of conduct:

<http://www.washington.edu/cssc/for-students/student-code-of-conduct/> I will make allowances for exceptional circumstances such as sickness, bereavement and official university business. I will not make exceptions for work, other classes, personal obligations, etc.

Any code or text that you copy from an external source must have a comment (or reference) indicating its origins.

Attendance: Attend all classes. You are responsible for all the material covered in class, as well as any announcements including change of due dates or assignment specifications. There will also be graded in-class group exercises to practice problem solving. If you miss a class, I expect you to make-up for it on your own by asking your friends, reviewing the textbook, lecture materials, etc.

Communication: We will use discord as an extension of the classroom. Use a meaningful nickname and act professionally. If your question can be answered publicly by me or by a classmate, post it to discord. Use the office hours for complex issues or topics you are struggling with.

Use your [UW email](#) rather than "Canvas Messaging" to communicate directly with me. "Canvas Submission Comments" should only be used to draw the grader's attention to a specific part of your submission.

Problems: If you are having difficulties, come and talk to me. If I don't know about it, I cannot help you. Small problems can be fixed easily early in the quarter, but might become impossible to fix later on.

Course Material: Lecture notes and other material will be posted to Canvas under "Files".

See the [School of STEM Course Policies](#), which covers:

- Academic Integrity
- Access and Accommodations
- Classroom Emergency Preparedness
- For Our Veterans
- Grade of Incomplete

- Inclement Weather
- Parenting Resources
- Religious Accommodations
- Respect for Diversity
- Student Support Services
- Surviving Sexual and Relationship Violence
- Wonder How to Address Faculty?

The UW Religious Accommodations Policy can be found at
<https://registrar.washington.edu/staffandfaculty/religious-accommodations-policy/>

Course Calendar

Topics and due dates below might be modified as needed.

See [Weekly Notes](#) for additional readings and pointers to external resources

Week	Tuesday/Thursday	Notes
	Week-1 Introduction, OS Structures	
	Week-2 Processes	
	Week-3 Threads, Scheduling	P1 : Unix Shell
	Week-4 Process Synchronization	P2: Sudoku
	Week-5 TBD	P3: Scheduling
	Week-6 Main Memory	Midterm
	Week-7 Virtual Memory	
	Week-8 File System	P4: Contiguous Memory Allocation
	Week-9 Deadlocks	
	Week-10 System Protection	Monday, 31 May, Memorial Day P5: BFS - A Unix-like File System
	Final Final - Thursday, 10 June	No classes during finals week